Econ 0100 | Classwork E3

Due in Recitation

Question 1 (of 1) | Big Flue Networks

Flue networks forms a convenient form of communication and transportation for those in the wizarding world. Its easy to move between flue portals on the network or use it to chat with those at a distance. Due to the benefits of adding *to* an existing network instead of starting a new one (i.e. network effects) there have historically only been two networks: Fast Flue Inc. and Soot Net. Suppose both networks have a constant marginal cost of 10 galleons, no fixed costs, and face a demand curve for flue network trips given by the following.

$$P = 200 - Q$$

The marginal revenue for Fast Flue Inc. is the following.

$$MR_F = 200 - 2q_F - q_S$$

The marginal revenue for Soot Net. is the following.

$$MR_S = 200 - 2q_S - q_F$$

The networks use a sophisticated ticketing system to charge a price for every trip. Prices are in galleons. This market is symmetric.

Q1.a | Fast Flue Inc.'s Best Response

What is the best response for Fast Flue Inc.?

Q1.b | Soot Net's Best Response

What is the best response for Soot Net?

Q1.c | Market Quantity

What is Nash equilibrium quantity for Fast Flue Inc. and Soot Net?

Q1.d | Equilibrium Price

What is the Nash equilibrium price in this market?

Q1.e | Market Profit

What is the total profit (sum of both firms profit) in this market?

Q1.e | Deadweight Loss

Graph and solve for the deadweight loss.